Dr. Abdelwaheb HAFS

Profile

PhD in Robotics and Human Movement Science, specialized in assistive control and human-robot interaction. Experience in control law design, predictive control, and robotics.

Contact

abdelwaheb.hafs@universite-paris-saclay.fr

abdelwahebhafs@gmail.com

linkedin.com/in/a-hafs

abdelwahebhafs.github.io

Paris, France

Skills

- **Programming Languages:** C, **C++, Python, MATLAB**, C#
- **Software:** LaTeX, Git, JASP, Inkscape, LabVIEW, SolidWorks
- **Teaching:** Control Systems, Mathematics, Physics, Robotics, Biomechanics, Programming

Languages

- French
- English

Interests

- Climbing
- Hiking
- Traveling
- Origami

EDUCATION

PhD in Robotics and Human Movement — *2021 – 2025*

Université Paris-Saclay (Orsay)

- **Dissertation:** Game-theoretic **predictive control** for intuitive movement assistance in human-robot interaction
- **Award:** DEMENY-VAUCANSON 2022 – Université Paris-Saclay

Master's Degree in Robotics, Assistance and Mobility — *2020 – 2021*

Université Paris-Saclay (Versailles)

- **Thesis: Adaptive control laws** to improve human-exoskeleton interaction
- Ranked **1st** out of 10 students

Double Degree: Engineering & Master's in Control Systems — *2017 – 2020*

École Nationale Polytechnique (Oran)

- **Thesis:** Predictive control applied to a two-wheeled mobile robot
- Ranked **2nd** out of 29 students

Preparatory Classes for Engineering Schools — *2015 – 2017*

École Supérieur en Génie Électrique et Énergétique (Oran)

- Ranked **3rd** out of 395 students

PROFESSIONAL EXPERIENCE

Temporary Teacher and Research Associate (ATER) — *2025 – 2026*

CIAMS Laboratory, Université Paris-Saclay (Orsay)

- **Research:** Implementation of intuitive assistive control laws with trajectory prediction for studying human-robot interaction
- **Teaching:** 192 hours in Biomechanics (L1 STAPS), **Robotics** (M1 ISMH), and **Programming** (Python) (M2 ISMH)

PhD Researcher — *2021 – 2025*

CIAMS Laboratory, Université Paris-Saclay (Orsay)

- **Design and implementation of control laws** based on differential game theory for intuitive human movement assistance using an exoskeleton robot (MATLAB, C++, Python)
- **Supervision of interns** within research projects
- **Teaching** (+200 hours): Biomechanics (L1 STAPS), **Robotics** (M1 ISMH), **Programming** (MATLAB) (M2 ISMH)
- PhD student representative, SSMMH Doctoral School
- Member of BOOST INRIA-CIAMS team

Final-Year Internship — *2021*

LURPA Laboratory, ENS Paris-Saclay (Gif-sur-Yvette)

- Synthesis of adaptive oscillator-based control for an exoskeleton robot arm (C++, Python)

Final-Year Engineering Project — *2020*

École Nationale Polytechnique (Oran)

- **Synthesis of predictive control** applied to a two-wheeled mobile robot (MATLAB, Simulink)

Other Projects

- **Presentations** at national and international conferences:
  IROS 2023 (Detroit, USA), MSR 2023 (Toulouse, France), BIOROB 2024 (Heidelberg, Germany), IFAC 2025 (Paris, France)
- Study of acceleration and jerk profiles for comfort analysis in transportation systems (Arduino, C++, MATLAB, LabVIEW), 2021